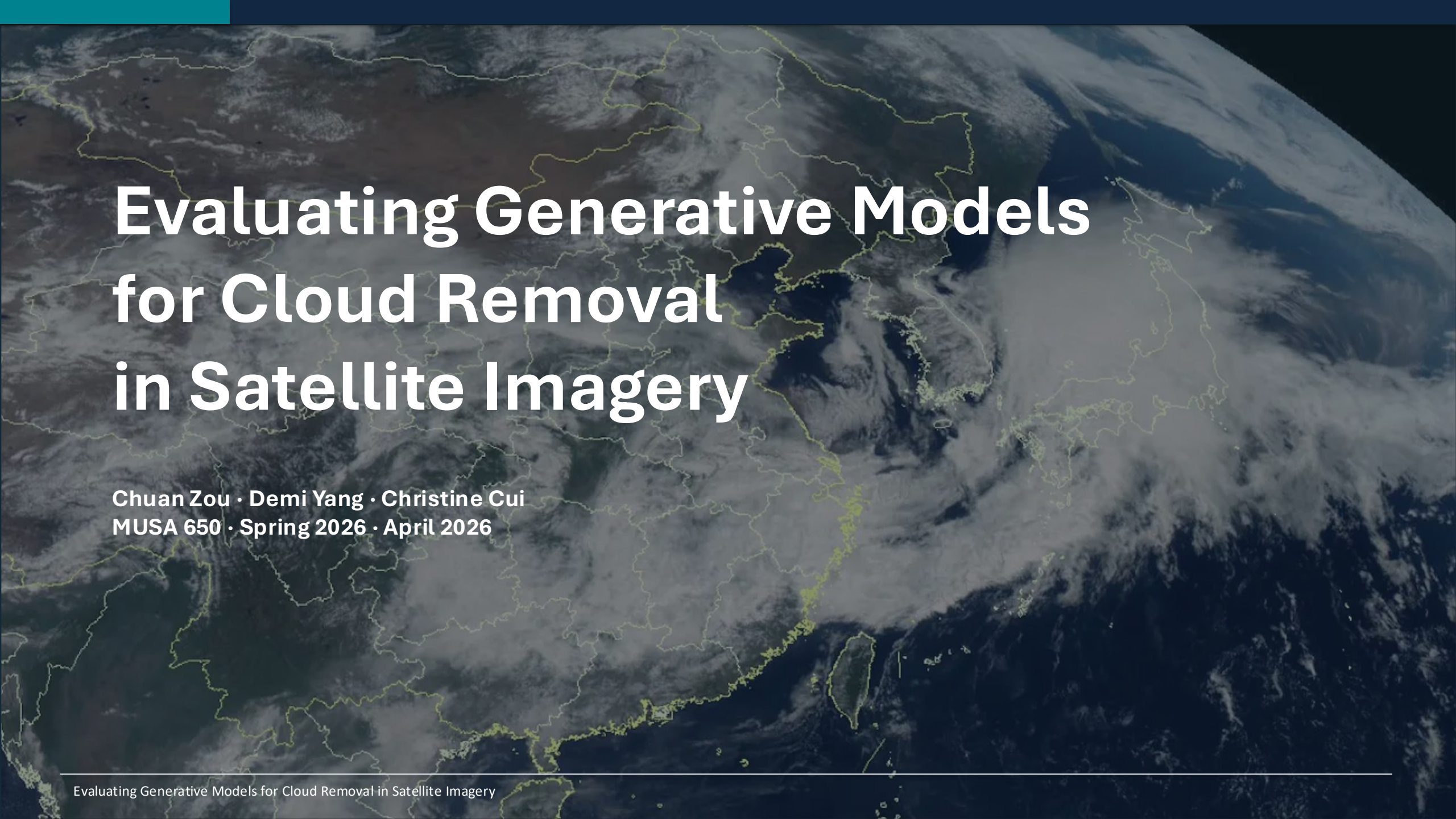


A satellite image of East Asia, showing the Korean Peninsula, Japan, and parts of China and the Philippines. The image is heavily overcast with white clouds. A thin yellow line outlines the coastlines and major landmasses. The title text is overlaid on the left side of the image.

Evaluating Generative Models for Cloud Removal in Satellite Imagery

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Why This Problem Matters

Motivating application: Yunnan and other cloud-prone southern regions



>80%

annual cloud frequency
in southern China

*Persistent cloud cover renders
optical satellites nearly useless*

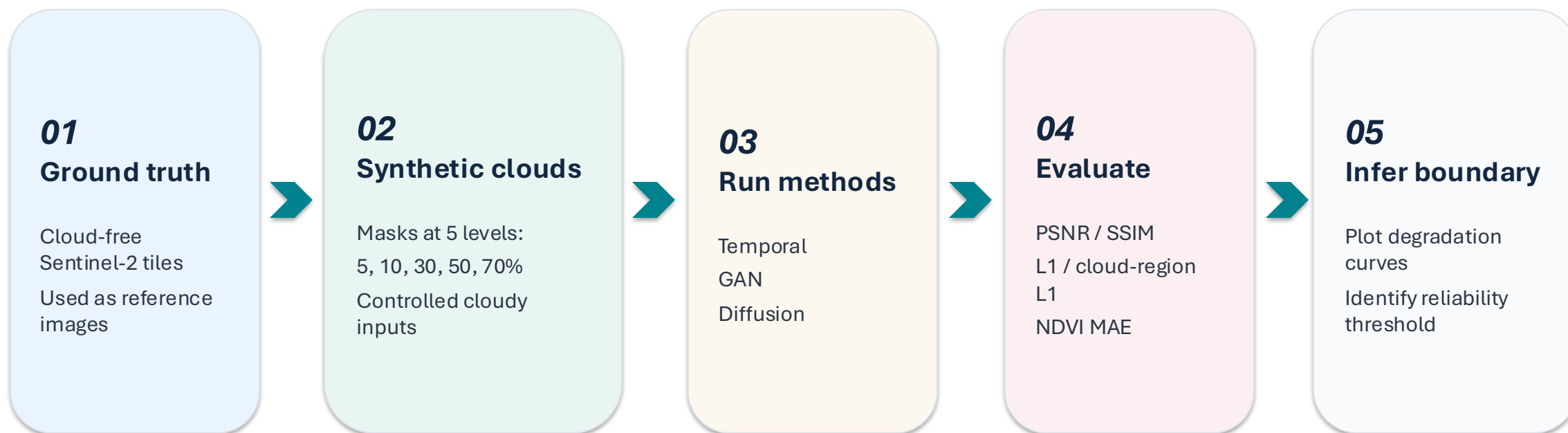
Research question :

"At what cloud coverage threshold do generative models become unreliable?"

Experimental Logic

A controlled benchmark lets us isolate the effect of cloud coverage.

We compare all methods on the **same scenes**, with the **same masks**, and at the **same five occlusion levels**.



Performance changes can be attributed directly to increasing cloud coverage rather than to arbitrary scene differences.

Data, Methods, and a Key Limitation

Important context for interpreting the benchmark.

Data

- Sentinel-2, 10 m resolution
- 2,000 cloud-free image tiles
- 70% train / 30% test split
- Input: RGB + NIR
- Output: RGB

Metrics

- PSNR
- SSIM
- L1 / cloud-region L1
- NDVI MAE

Methods

- Temporal compositing (baseline): Uses historical cloud-free pixels as a non-deep-learning baseline.
- GAN-based cloud removal: Learns a direct cloudy-to-clear image translation
- Diffusion-based inpainting: Gradually reconstructs masked regions through denoising steps

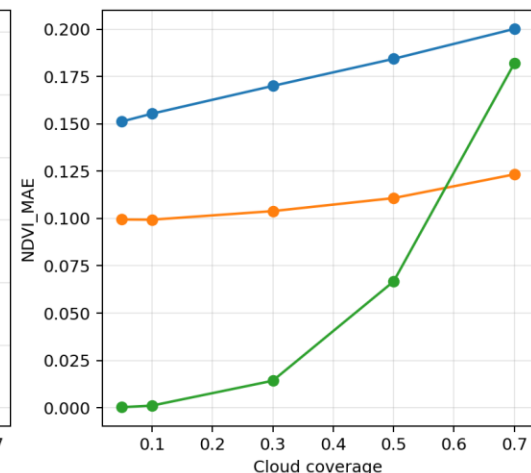
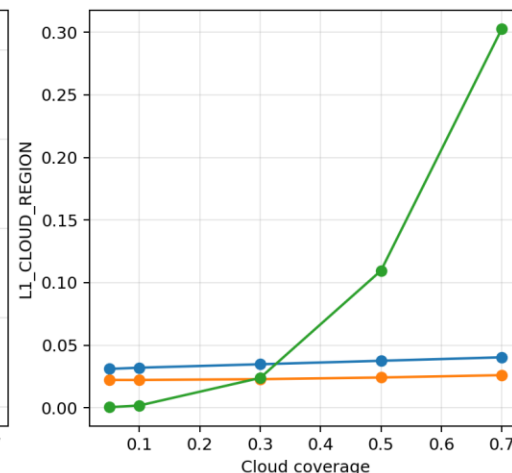
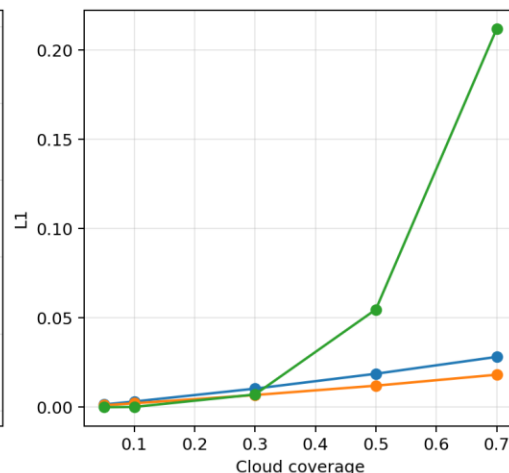
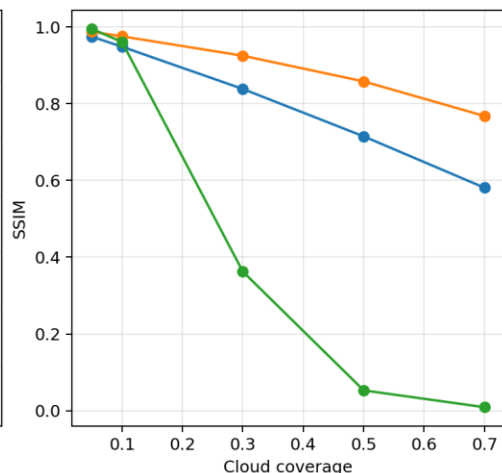
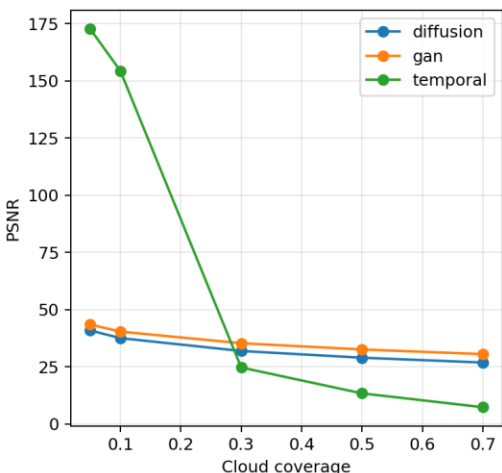
Training setup: Batch size = 8, Epochs = 5, Learning rate = 0.0002

Important limitation: the data are selected in Beijing, but the motivating application is Yunnan.

This benchmark is stronger for studying occlusion sensitivity than for claiming direct generalization to Yunnan.

Quantitative Comparison Across Cloud Coverage Levels

5



PSNR
Pixel accuracy
higher is better

SSIM
Structural similarity
higher is better

L1
Overall error
lower is better

L1 in cloud region
Error in masked area
lower is better

NDVI MAE
Vegetation error
lower is better

Readout 1

GAN is best overall

Best balance of fidelity, structure, and vegetation accuracy as coverage rises.

Readout 2

Temporal is strong only at low coverage

Excellent at 5%–10%, but collapses after ~30% when too much information is missing.

Readout 3

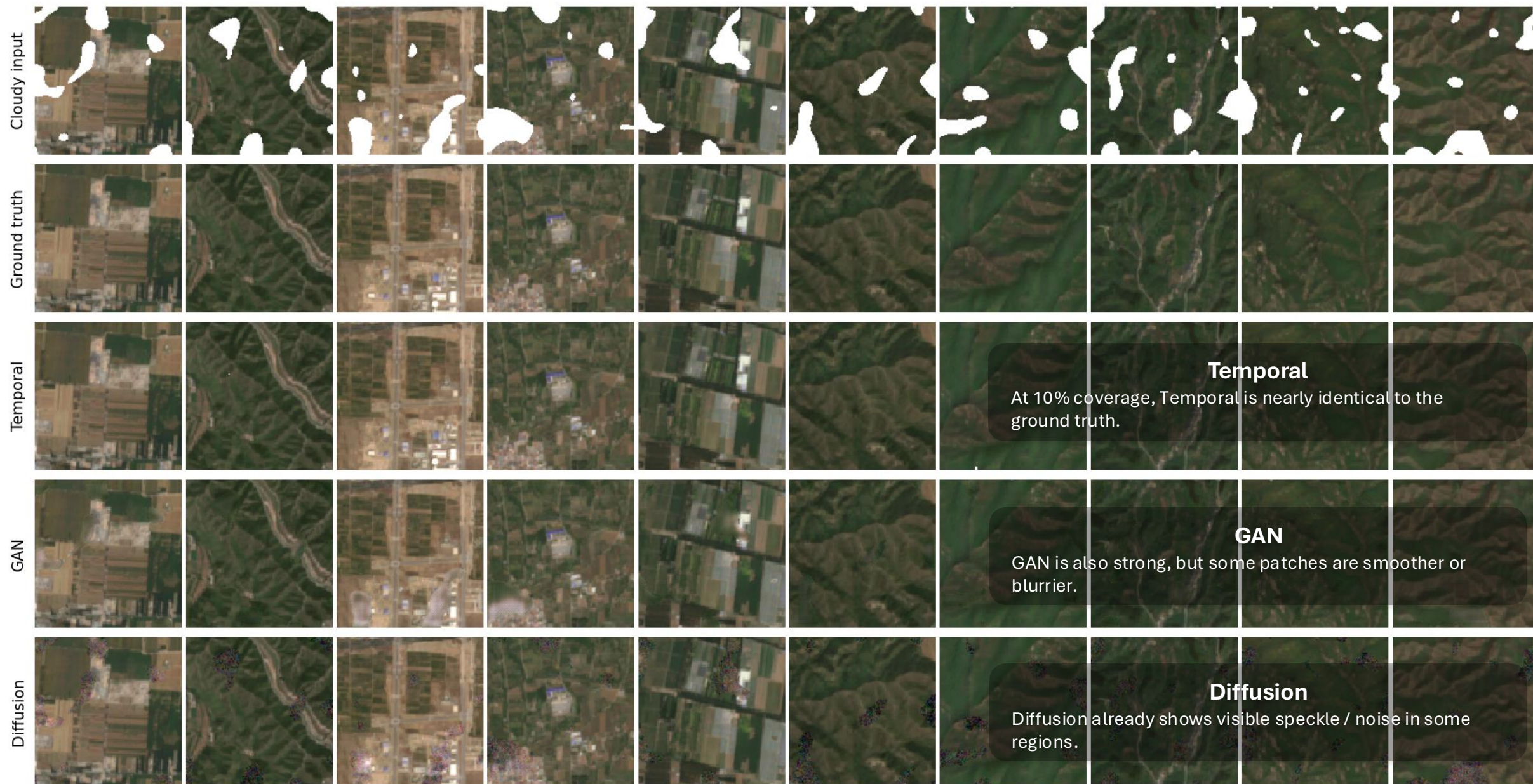
Diffusion underperforms here

In this setup, it degrades steadily and is not competitive with GAN.

PSNR and SSIM should go up; L1, cloud-region L1, and NDVI MAE should go down.

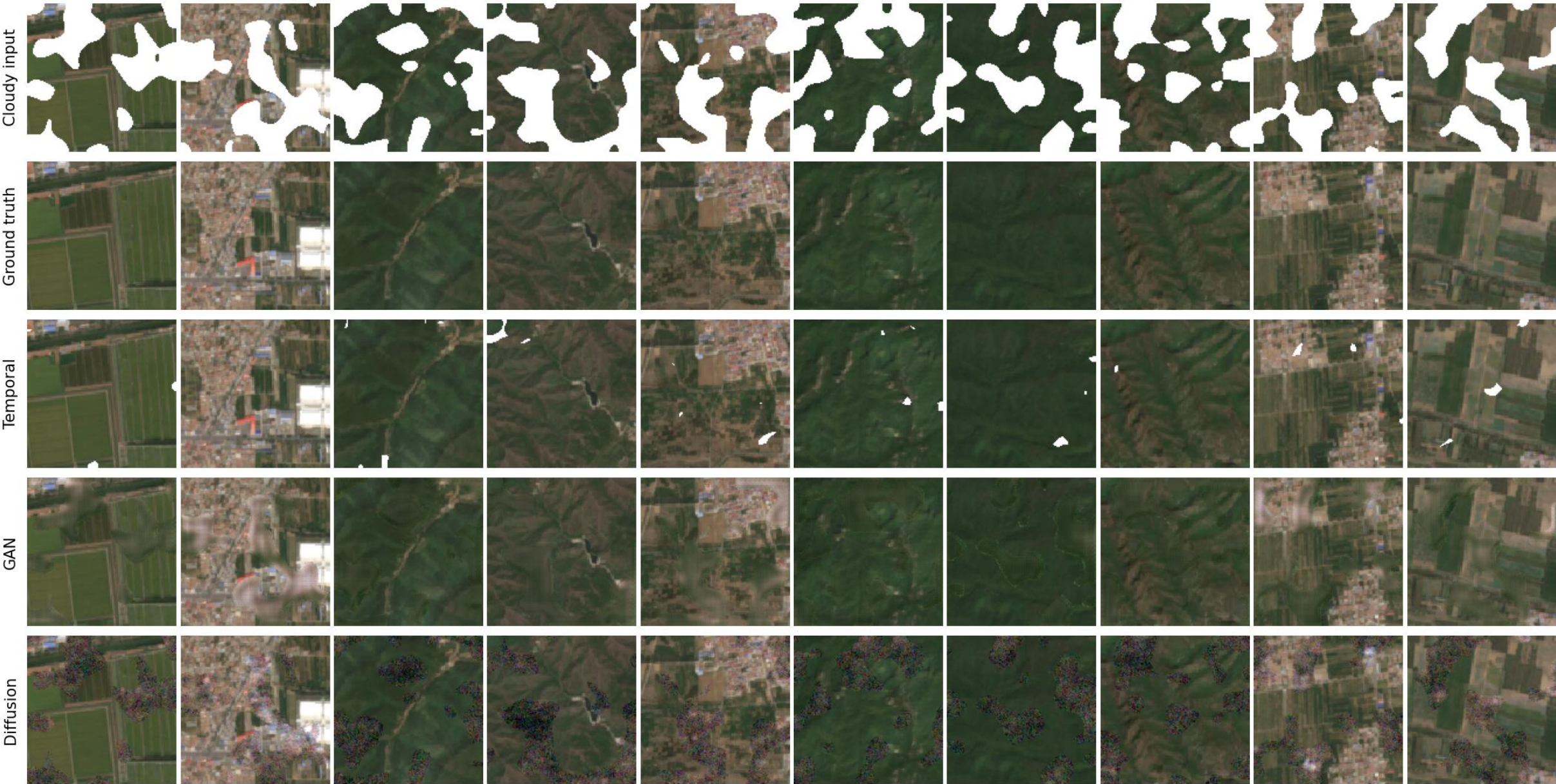
Qualitative Example at Low Coverage (10%)

The task is solvable in easier cases



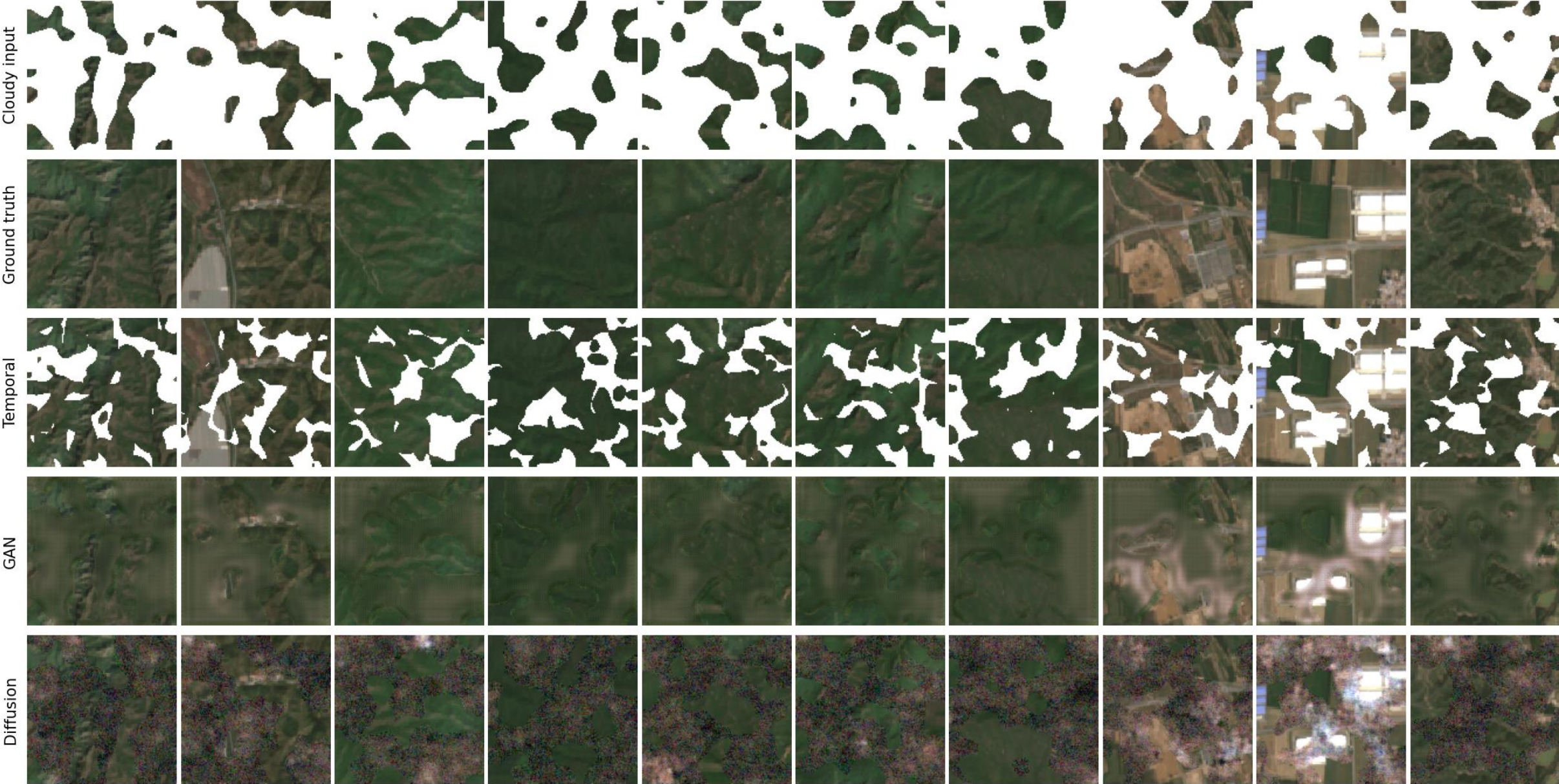
Qualitative Example at Low Coverage (30%)

Qualitative comparison @ coverage = 0.30

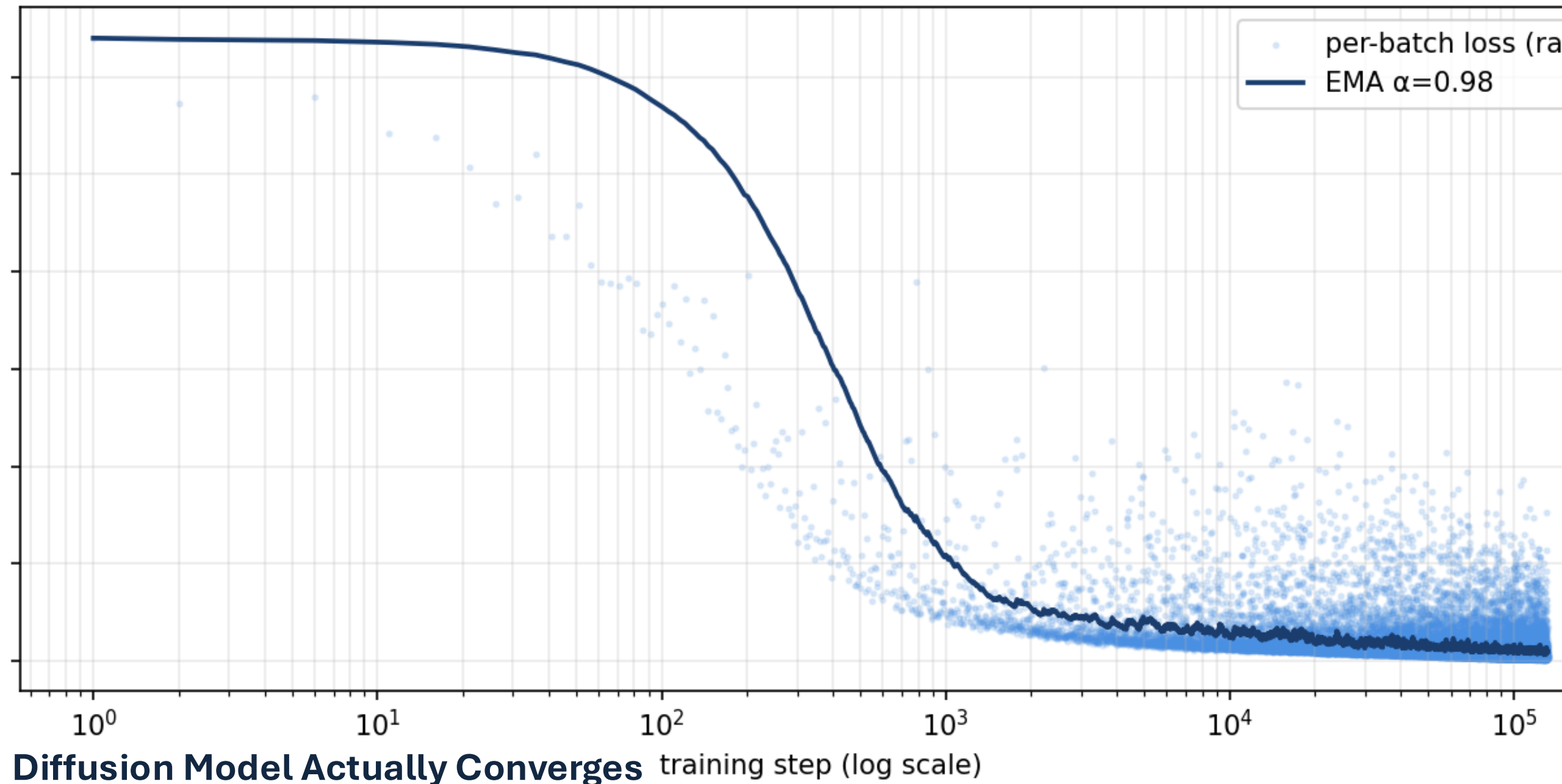


Qualitative Example at Low Coverage (70%)

Qualitative comparison @ coverage = 0.70



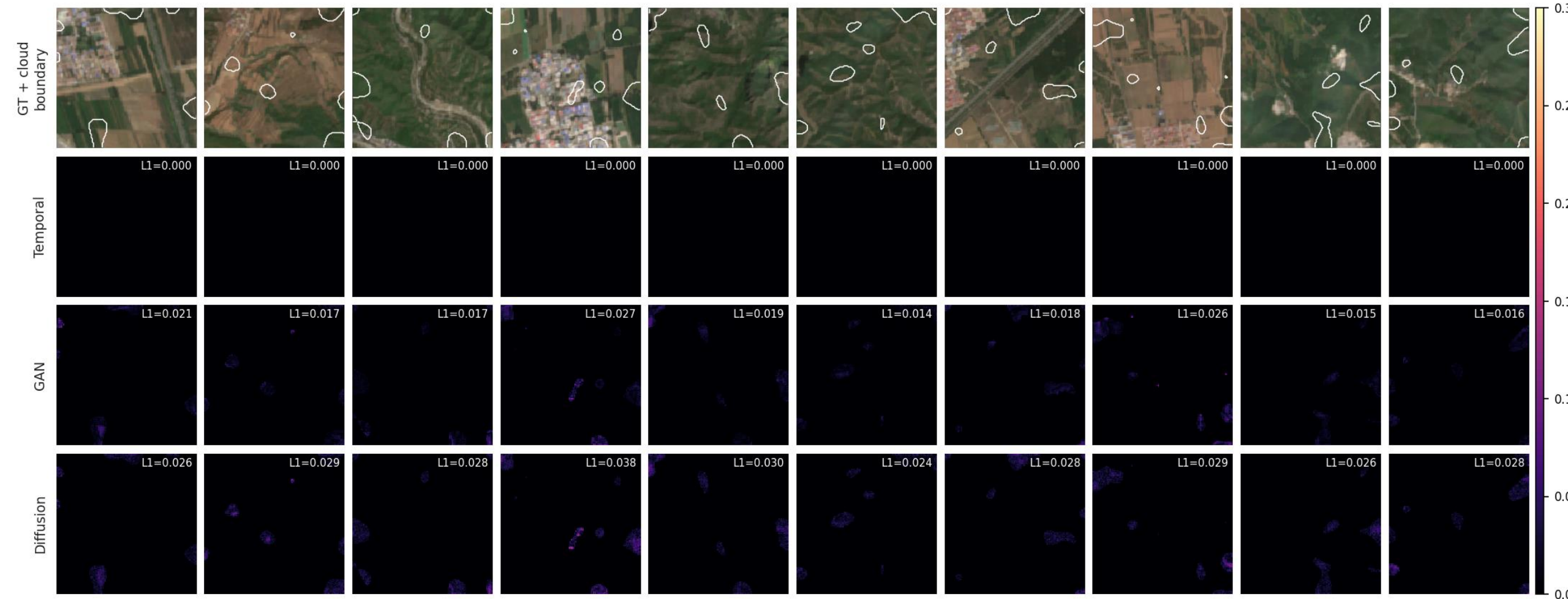
Diffusion training loss vs step



Diffusion Model Actually Converges

Per-pixel L1 Error Heatmap at High Coverage (5%)

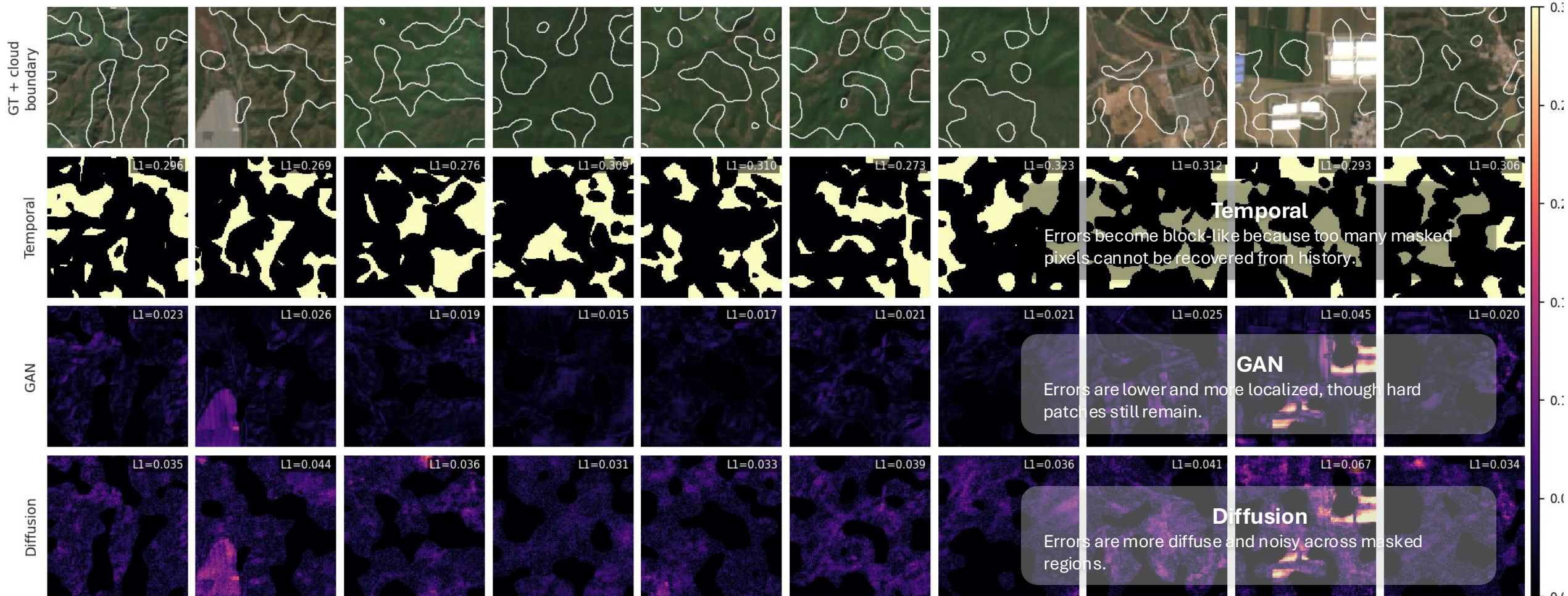
Per-pixel L1 error @ coverage = 0.05



Per-pixel L1 Error Heatmap at High Coverage (70%)

7

Hard case: spatial failure modes



The high errors of some patches are particularly concentrated in: White or highly reflective building areas Urban/industrial plots Areas with drastic texture changes. This indicates that under high occlusion, the model is more prone to making incorrect corrections in these complex areas.

Main findings

- Reliability depends strongly on cloud coverage.
- Temporal compositing is excellent at low coverage, but breaks down as missing area grows.
- GAN is the strongest overall method in this benchmark.

Limitations and next steps

- The benchmark uses Beijing tiles, while the motivating application is Yunnan / southern China.
- So the results are more convincing about occlusion sensitivity than about cross-region transfer.
- Next step: test on Yunnan-like or multi-region data to validate whether the same threshold still holds.

Why Did Diffusion Underperform?

1. Limited data and compute: Diffusion models are more data- and compute-intensive than GAN.
2. Simple backbone design: The current backbone (UNet) is too simple for complex remote-sensing textures.
3. Non-targeted optimization: The loss does not emphasize cloud-region recovery enough.
4. Weak mask conditioning: The model does not focus strongly enough on the masked cloud regions.